BTECHBUDDY - DSA Previous Year Solved Papers

Semester Exam Solved Question Bank (2020 - 2024)

University: AKTU, RTU, VTU, GGSIPU Standard Curriculum

Section A: 2-Marks Short Answer Questions Solved

Q1: Define Balance Factor of an AVL tree node. Ans: HeightLeft - HeightRight

Q2: Differentiate between Stack and Queue. Ans: LIFO vs FIFO principle.

Section B: 7-Marks Long Answer Questions with Full Stepwise Derivation

Q3: Construct AVL tree for keys: 10, 20, 30, 40, 50, 25 with rotational diagrams.

Q4: Write iterative algorithm to reverse a singly linked list in On time and O1 space.

Section C: 10-Marks Code Implementations and Complexities

Q5: Trace Dijkstra Algorithm on a 6-node weighted graph step by step.

Verified by University Gold Medalists. BTECHBUDDY Academic Team.